

Effective Search Techniques

When searching for materials in electronic resources and databases, utilizing specific search strategies helps find useful information quickly and avoids wasting time gathering irrelevant details.

1. Keyword Searching

- Uses "natural language" by entering relevant words or phrases directly into a search box.
- The engine searches all parts of a source for the entered words.

2. Boolean Searching

A logic-based search technique that uses operators to narrow down or broaden search results with more precision. Operators must be capitalized and can be used multiple times in a single search.

- **AND:** Connects words or phrases; **both** terms must appear somewhere in the search results (e.g., gender AND Shakespeare).
- **OR:** Used to look up synonyms or phrases that mean nearly the same thing; broadens results (e.g., car OR automobile).
- **NOT:** Excludes specific words or phrases from the search results (e.g., "video games" NOT teenagers).
- **Combining Operators:** Multiple operators can be combined for specific searches (e.g., "animal cloning" OR "animal duplication" AND jellyfish NOT sheep).

3. Other Search Enhancements

- **Limiters:** Drop-down menus or checkboxes on search pages that restrict results by criteria like date of publication, material type, or full text.
- **Subject Searching:** Searching by pre-defined categories found within the subject field of an item record.
- **Phrase Searching:** Putting quotation marks around a phrase (e.g., "corporate social responsibility") forces the engine to return results containing those exact words in that exact order.
- **Using References/Works Cited Lists:** Following bibliographies or lists of works cited in a good book or article to locate additional valuable sources.

Understanding and Comparing Search Engines

A search engine is a software program that continuously scans and indexes internet pages to help users find information quickly using keywords or phrases.

Overview of Popular Search Engines

- **Google:** The largest search engine in the world. Developed in 1996 by Larry Page and Sergei Brin. It features a highly personalized, AI-driven experience with full search history control via "My Activity," but collects extensive data for ad targeting.
- **Bing:** Owned by Microsoft (formerly Live Search). It offers highly personalized results and extensive tracking similar to Google, but includes a prominent rewards program for web browsing.
- **Yahoo:** An internet portal and media company that features a search engine powered by Bing. It provides a web-portal style interface integrated with services like Yahoo Mail and Finance.
- **DuckDuckGo:** A privacy-focused search engine that **does not** track search history, personalize results, or target ads based on personal data.
- **Brave Search:** The default engine for the Brave Browser (usable on others). It tracks nothing and, unlike DuckDuckGo, uses its own proprietary index to ensure it is never forced by larger engines to include hidden trackers.
- **Ask.com:** Utilizes an automated algorithm called "ExpertRank," which orders results based on topic communities and expertise rather than pure popularity.
- **Baidu & Yandex:** Main search engines dominating specific regions—Baidu in China (over 600 million users) and Yandex in Russia/Russophone countries.

Comparison Matrix at a Glance

Feature	Google	Bing	Yahoo	DuckDuckGo
Search Results	Most accurate, AI-driven, personalized.	Most accurate, AI-driven, personalized.	Powered by Bing; similar results.	Relevant but less comprehensive than Google.
Privacy & Data	Collects extensive user data for ads.	Collects extensive user data for ads.	Collects data for ads, but less aggressive than Google.	Does not track users or store personal data.
History Management	Full control over search history/activity via dashboard.	Full control over search history and activity.	History management via Yahoo account.	No search history tracking or personalization.
Interface Style	Clean, minimalist, frequently updated.	Clean, minimalist, frequently updated.	Web portal style integrated with news/services.	Simple, clean, privacy-focused.